

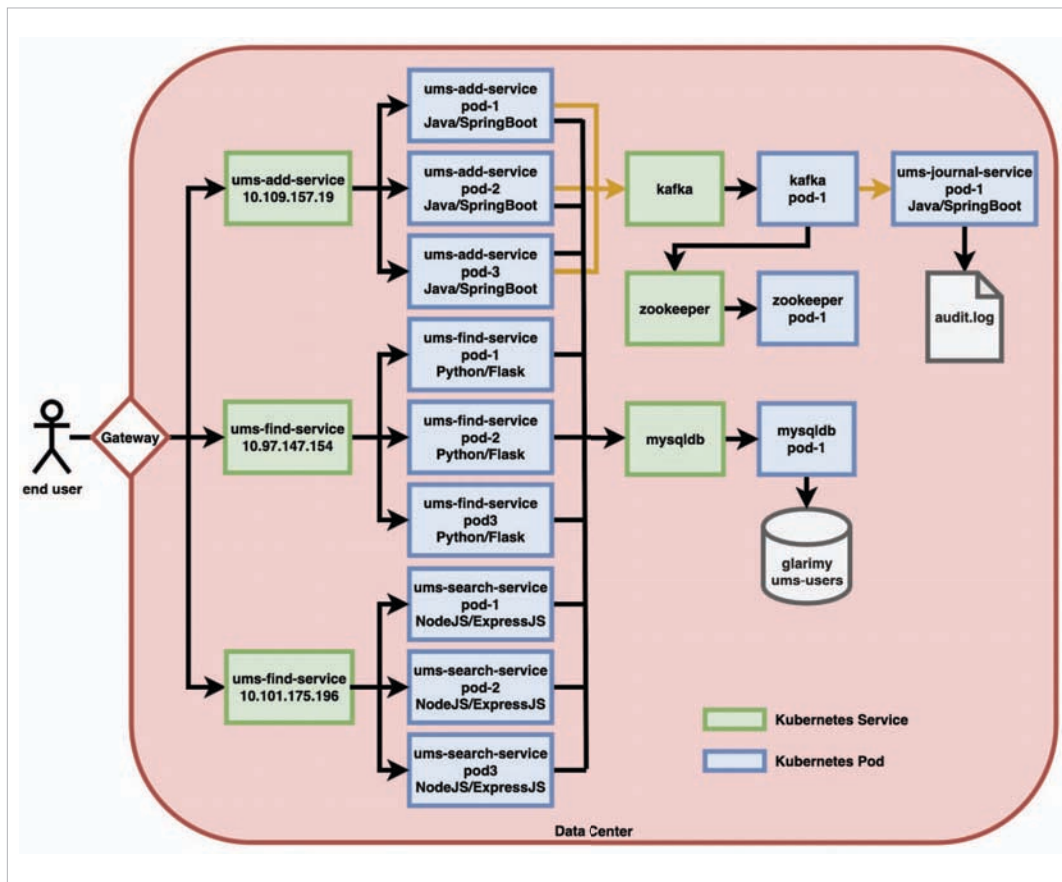


Figure 5: Final architecture of UMS

But how do you build such a gateway? Well, in most of the cases you will use third-party ready-to-use gateway solutions like Kong, etc. The non-trivial gateways offer a lot of other features besides simple request delegation. They do filtering, throttling, protocol-conversion, security, and a lot of other things.

And other patterns

Before closing, let us also have a look at a few other microservices patterns.

Service discovery is one such pattern. Entities like *gateway* and others are required to discover the service instances that are deployed on the cluster. A *service registry* is used to maintain the data of all the services at a central location. Service may be registered in two different ways: 1) The service itself registers with the *service registry*. This is called the *self-registration pattern*; 2) The deployer registers the service. This is called the *registrar pattern*. In either case, once the service is registered, others will be able to discover it.

And the last set of patterns we deal with falls under the *observability* aspect. Unless there is a mechanism to know the health of the services and the state of the requests, it

is impossible to guarantee reliability. Many platforms offer distributed tracing, aggregated logs, application metrics, etc, to aid observability.

Epilogue

To sum up, the microservices architecture has been well-tried and well-documented since it captured the imagination of the industry about a decade ago. A designer with a grasp of the patterns (ranging from basic object-oriented patterns, domain-driven patterns, messaging

patterns, and distributed patterns to microservices patterns) and a hand on a few tools that implement these patterns is well positioned to architect web-scale mission-critical applications.

In this *Design Odyssey* series, we covered some nuances of designing modern applications over 12 parts. Though we used a simple user management system to illustrate some critical patterns, a good understanding of these concepts and patterns is sufficient even when dealing with a complex system.

The code base of the series is available at <https://bitbucket.org/glarimy/the-design-odyssey>.

A list of references and other resources will also be added when this series takes the form of a book. END 🐧

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